

3741. Minimum Distance Between Three Equal Elements II

Solved 

Medium

 Topics

 Companies

 Hint

You are given an integer array `nums`.

A tuple (i, j, k) of 3 **distinct** indices is **good** if `nums[i] == nums[j] == nums[k]`.

The **distance** of a **good** tuple is `abs(i - j) + abs(j - k) + abs(k - i)`, where `abs(x)` denotes the **absolute value** of `x`.

Return an integer denoting the **minimum** possible **distance** of a **good** tuple. If no **good** tuples exist, return `-1`.

Example 1:

Input: `nums = [1,2,1,1,3]`

Output: 6

Explanation:

The minimum distance is achieved by the good tuple $(0, 2, 3)$.

$(0, 2, 3)$ is a good tuple because `nums[0] == nums[2] == nums[3] == 1`. Its distance is `abs(0 - 2) + abs(2 - 3) + abs(3 - 0) = 2 + 1 + 3 = 6`.

Example 2:

Input: `nums = [1, 1, 2, 3, 2, 1, 2]`

Output: 8

Explanation:

The minimum distance is achieved by the good tuple `(2, 4, 6)`.

`(2, 4, 6)` is a good tuple because `nums[2] == nums[4] == nums[6] == 2`. Its distance is `abs(2 - 4) + abs(4 - 6) + abs(6 - 2) = 2 + 2 + 4 = 8`.

Example 3:

Input: `nums = [1]`

Output: -1

Explanation:

There are no good tuples. Therefore, the answer is -1.

Constraints:

- `1 <= n == nums.length <= 105`
- `1 <= nums[i] <= n`

Python:

```
from collections import defaultdict
```

```
class Solution:
```

```
    def minimumDistance(self, nums):
```

```
        mp = defaultdict(list)
```

```
        for i, num in enumerate(nums):
```

```
            mp[num].append(i)
```

```
        mini = float('inf')
```

```
        for temp in mp.values():
```

```
            m = len(temp)
```

```
            if m >= 3:
```

```
                for i in range(m - 2):
```

```

a = temp[i]
b = temp[i + 1]
c = temp[i + 2]

diff = 2 * (max(a, b, c) - min(a, b, c))
mini = min(mini, diff)

```

return -1 if mini == float('inf') else mini

JavaScript:

```

var minimumDistance = function (nums) {
  const pos = new Map()
  for (let i = 0; i < nums.length; i++) {
    if (!pos.has(nums[i])) {
      pos.set(nums[i], [])
    }
    pos.get(nums[i]).push(i)
  }
  let ans = Infinity
  for (const [val, indices] of pos.entries()) {
    if (indices.length >= 3) {
      for (let i = 0; i <= indices.length - 3; i++) {
        ans = Math.min(ans, 2 * (indices[i + 2] - indices[i]))
      }
    }
  }
  return ans === Infinity ? -1 : ans
};

```

Java:

```

import java.util.*;

class Solution {
  public int minimumDistance(int[] nums) {
    Map<Integer, List<Integer>> mp = new HashMap<>();
    int n = nums.length;

    for (int i = 0; i < n; i++) {
      mp.computeIfAbsent(nums[i], k -> new ArrayList<>()).add(i);
    }

    int mini = Integer.MAX_VALUE;

    for (List<Integer> temp : mp.values()) {
      int m = temp.size();
    }
  }
}

```

```
    if (m >= 3) {  
        for (int i = 0; i < m - 2; i++) {  
            int a = temp.get(i);  
            int b = temp.get(i + 1);  
            int c = temp.get(i + 2);  
  
            int diff = 2 * (Math.max(a, Math.max(b, c)) - Math.min(a, Math.min(b, c)));  
            mini = Math.min(mini, diff);  
        }  
    }  
}  
  
return mini == Integer.MAX_VALUE ? -1 : mini;  
}
```